

Representative Structural Calculation Package

U-shaped school wing

Item	Value
Project	U-shaped school wing
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	03-u-shaped-school

Nodes	Members	Loads	Combos	Max disp.	Max util.
40	57	174	28	63.66 mm	4.301

Model And Design Basis

Item	Value
Building type	school
Description	Three-story U-shaped education building around an open courtyard.
Occupancy	School
Bounds X/Y/Z	21.00 / 18.00 / 11.10
Analysis status	OK
Design status	OK
Package audit	OK

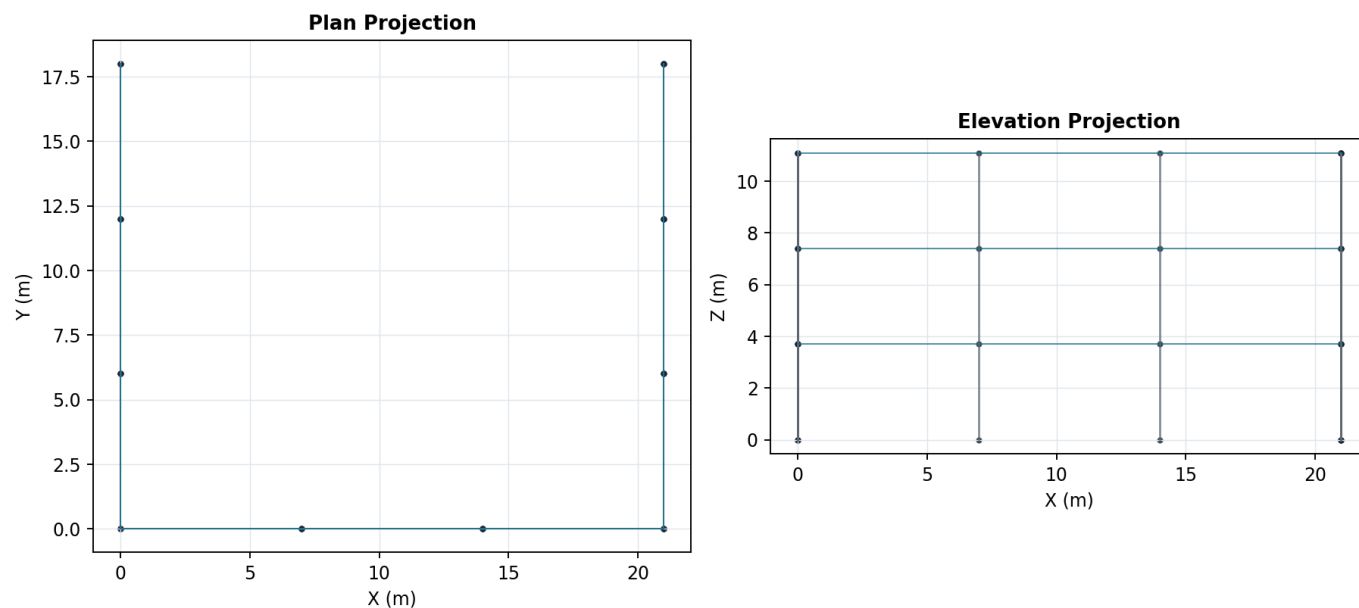
Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	27	0.000 kN	0.000 kN	-4309 kN
L	live	27	0.000 kN	0.000 kN	-2079 kN
WX	wind	30	89.91 kN	0.000 kN	0.000 kN
WY	wind	30	0.000 kN	104.89 kN	0.000 kN
EX	seismic	30	352.51 kN	0.000 kN	0.000 kN
EY	seismic	30	0.000 kN	352.51 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	378.00	1436 kN	831.60 kN	29.97 kN	34.97 kN
2	378.00	1436 kN	831.60 kN	29.97 kN	34.97 kN
3	378.00	1436 kN	415.80 kN	29.97 kN	34.96 kN

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Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	3.438 mm	0.403	0.000
KDS-ST-02	OK	4.248 mm	0.516	0.000
KDS-ST-04-WX-P	OK	13.69 mm	0.463	0.000
KDS-ST-04-WX-N	OK	13.69 mm	0.463	0.000
KDS-ST-04-WY-P	OK	5.882 mm	0.424	0.000
KDS-ST-04-WY-N	OK	5.431 mm	0.462	0.000
KDS-ST-05-EX-P	OK	63.66 mm	0.728	0.000
KDS-ST-05-EX-N	OK	63.66 mm	0.728	0.000
KDS-ST-05-EY-P	OK	20.03 mm	0.475	0.000
KDS-ST-05-EY-N	OK	19.51 mm	0.588	0.000
KDS-ST-06-WX-P	OK	13.40 mm	0.281	0.000
KDS-ST-06-WX-N	OK	13.40 mm	0.281	0.000
KDS-ST-06-WY-P	OK	5.227 mm	0.247	0.000
KDS-ST-06-WY-N	OK	4.937 mm	0.273	0.000
KDS-ST-07-EX-P	OK	63.38 mm	0.632	0.000
KDS-ST-07-EX-N	OK	63.38 mm	0.632	0.000
KDS-ST-07-EY-P	OK	19.77 mm	0.358	0.000

KDS-ST-07-EY-N	OK	19.46 mm	0.403	0.000
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Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M29	NG	4.301	steel-deflection	KDS-ST-05-EX-P
2	M30	NG	4.301	steel-deflection	KDS-ST-05-EX-N
3	M27	NG	3.369	steel-deflection	KDS-ST-05-EY-N
4	M28	NG	3.369	steel-deflection	KDS-ST-05-EY-N
5	M56	NG	2.652	steel-deflection	KDS-ST-05-EY-N
6	M57	NG	2.652	steel-deflection	KDS-ST-05-EY-N
7	M19	NG	2.399	steel-deflection	KDS-ST-05-EX-P
8	M20	NG	2.399	steel-deflection	KDS-ST-05-EX-N
9	M25	NG	2.084	steel-deflection	KDS-ST-05-EY-P
10	M26	NG	2.084	steel-deflection	KDS-ST-05-EY-P
11	M54	NG	2.077	steel-deflection	KDS-ST-05-EX-N
12	M55	NG	2.077	steel-deflection	KDS-ST-05-EX-P
13	M17	NG	1.860	steel-deflection	KDS-ST-05-EX-N
14	M18	NG	1.860	steel-deflection	KDS-ST-05-EX-P
15	M47	NG	1.480	steel-deflection	KDS-ST-05-EY-N
16	M48	NG	1.480	steel-deflection	KDS-ST-05-EY-N
17	M50	NG	1.285	steel-deflection	KDS-ST-05-EX-P
18	M53	NG	1.285	steel-deflection	KDS-ST-05-EX-N
19	M22	NG	1.224	steel-deflection	KDS-ST-05-EX-P
20	M23	NG	1.224	steel-deflection	KDS-ST-05-EX-N

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	57	41	4.301
Connections	57	34	4.299
Foundations	10	34	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.
- Full KDS equation trace for every generated load combination and project-specific exception.

- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.